```
import tensorflow.compat.v1 as tf
tf.disable_v2_behavior()
```

BTP CODE 1.0

```
Start coding or generate with AI.
```

```
from google.colab import drive
drive.mount('/content/drive')

import os
os.makedirs('/content/drive/MyDrive/NGCF_weights', exist_ok=True)
print("✅ Drive ready!")
```

```
Mounted at /content/drive
✅ Drive ready!
```

```
#don't run this
#import tensorflow as tf
#print(tf.__version__)
```

```
2.19.0
```

```
!git clone https://github.com/xiangwang1223/neural_graph_collaborative_filtering.git
%cd neural_graph_collaborative_filtering
```

```
fatal: destination path 'neural_graph_collaborative_filtering' already exists and is not an empty directory.
/content/neural_graph_collaborative_filtering/NGCF/neural_graph_collaborative_filtering
```

```
!pip install -q numpy scipy torch
```

```
import torch
print("GPU available:", torch.cuda.is_available())
print("GPU:", torch.cuda.get_device_name(0) if torch.cuda.is_available() else "CPU")
```

```
GPU available: True
GPU: Tesla T4
```

```
#import os
print(os.listdir("Data"))
```

```
['README', 'gowalla', 'amazon-book']
```

```
print(os.listdir("Data/amazon-book"))
```

```
['test.txt', 'user_list.txt', 'train.txt', 'item_list.txt', 'README.md']
```

```
%cd /content/neural_graph_collaborative_filtering
```

```
/content/neural_graph_collaborative_filtering
```

```
!ls
```

```
Data  LICENSE  NGCF  README.md
```

```
%cd NGCF
```

```
/content/neural_graph_collaborative_filtering/NGCF
```

```
!ls
```

```
BPRMF.py  NGCF.py  NMF.py  README.md  utility
```

```
with open('/content/neural_graph_collaborative_filtering/NGCF/NGCF.py', 'r') as f:
    content = f.read()

# Fix 1: TF compatibility
content = content.replace(
    'import tensorflow as tf',
    'import tensorflow.compat.v1 as tf\ntf.disable_v2_behavior()'
)

# Fix 2: xavier initializer
content = content.replace(
```

```
    'tf.contrib.layers.xavier_initializer()',
    'tf.glorot_uniform_initializer()'
)

# Fix 3: other contrib.layers references
content = content.replace(
    'tf.contrib.layers.',
    'tf.layers.'
)

# Fix 4: np.mat removed in NumPy 2.0
content = content.replace(
    'np.mat([coo.row, coo.col]).transpose()',
    'np.asmatrix(np.vstack([coo.row, coo.col]).T)'
)

with open('/content/neural_graph_collaborative_filtering/NGCF/NGCF.py', 'w') as f:
    f.write(content)

print("All patches applied!")

# Verify no remaining issues
issues = {'contrib': [], 'np.mat': []}
with open('/content/neural_graph_collaborative_filtering/NGCF/NGCF.py', 'r') as f:
    lines = f.readlines()

for i, line in enumerate(lines, 1):
    if 'contrib' in line:
        issues['contrib'].append(f"  Line {i}: {line.strip()}")
    if 'np.mat' in line and 'asmatrix' not in line:
        issues['np.mat'].append(f"  Line {i}: {line.strip()}")

for issue_type, found in issues.items():
    if found:
        print(f"⚠️  Remaining {issue_type} references:")
        for f in found:
            print(f)
    else:
        print(f"✅  No {issue_type} issues found")

with open('/content/neural_graph_collaborative_filtering/NGCF/NGCF.py', 'r') as f:
    content = f.read()

content = content.replace('import tensorflow as tf',
    'import tensorflow.compat.v1 as tf\ntf.disable_v2_behavior()')
content = content.replace('tf.contrib.layers.xavier_initializer()',
    'tf.glorot_uniform_initializer()')
content = content.replace('tf.contrib.layers.', 'tf.layers.')
content = content.replace('np.mat([coo.row, coo.col]).transpose()',
    'np.asmatrix(np.vstack([coo.row, coo.col]).T)')

with open('/content/neural_graph_collaborative_filtering/NGCF/NGCF.py', 'w') as f:
    f.write(content)

# ✅ NEW: Fix metrics.py
with open('/content/neural_graph_collaborative_filtering/NGCF/utility/metrics.py', 'r') as f:
    metrics_content = f.read()

metrics_content = metrics_content.replace(
    'np.asfarray',
    'np.asarray'
)

with open('/content/neural_graph_collaborative_filtering/NGCF/utility/metrics.py', 'w') as f:
    f.write(metrics_content)

print("✅ All patches applied — NGCF.py + metrics.py!")
```

```
All patches applied!
✅  No contrib issues found
✅  No np.mat issues found
✅ All patches applied — NGCF.py + metrics.py!
```

```
with open('NGCF.py', 'r') as f:
    content = f.read()

content = content.replace('import tensorflow as tf',
    'import tensorflow.compat.v1 as tf\ntf.disable_v2_behavior()')
content = content.replace('tf.contrib.layers.xavier_initializer()',
    'tf.glorot_uniform_initializer()')
content = content.replace('tf.contrib.layers.', 'tf.layers.')
content = content.replace('np.mat([coo.row, coo.col]).transpose()',
    'np.asmatrix(np.vstack([coo.row, coo.col]).T)')
```

```
with open('NGCF.py', 'w') as f:
    f.write(content)
print("✅ Patched!")
```

```
✅ Patched!
```

```
# Check if any contrib references remain
with open('/content/neural_graph_collaborative_filtering/NGCF/NGCF.py', 'r') as f:
    lines = f.readlines()

for i, line in enumerate(lines, 1):
    if 'contrib' in line:
        print(f"Line {i}: {line.strip()}")

print("Done checking — if nothing printed above, you're clean!")
```

```
Done checking — if nothing printed above, you're clean!
```

```
#!python NGCF.py --dataset amazon-book --epoch 200
files = os.listdir('/content/drive/MyDrive/NGCF_weights/')

if files:
    print(f"✅ Found saved weights: {files}")
    print("▶️  Resuming training...")
    !python NGCF.py \
        --dataset amazon-book \
        --epoch 20 \
        --save_flag 1 \
        --pretrain 1 \
        --weights_path /content/drive/MyDrive/NGCF_weights/ \
        --regs [1e-5]
else:
    print("❌ No weights found, starting fresh...")
    !python NGCF.py \
        --dataset amazon-book \
        --epoch 20 \
        --save_flag 1 \
        --weights_path /content/drive/MyDrive/NGCF_weights/ \
        --regs [1e-5]
```

```
✅ Found saved weights: ['weights']
▶️  Resuming training...
2026-03-31 18:59:45.256215: E external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:467] Unable to register cuFFT factory: Attempting to regi
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
E0000 00:00:1774983585.288983   33471 cuda_dnn.cc:8579] Unable to register cuDNN factory: Attempting to register factory for plugin cuDNN when
E0000 00:00:1774983585.300351   33471 cuda_blas.cc:1407] Unable to register cuBLAS factory: Attempting to register factory for plugin cuBLAS wh
W0000 00:00:1774983585.327143   33471 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking
W0000 00:00:1774983585.327180   33471 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking
W0000 00:00:1774983585.327188   33471 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking
W0000 00:00:1774983585.327195   33471 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking
WARNING:tensorflow:From /usr/local/lib/python3.12/dist-packages/tensorflow/python/compat/v2_compat.py:98: disable_resource_variables (from tens
Instructions for updating:
non-resource variables are not supported in the long term
n_users=52643, n_items=91599
n_interactions=2984108
n_train=2380730, n_test=603378, sparsity=0.00062
already load adj matrix (144242, 144242) 0.5650155544281006
use the normalized adjacency matrix
using xavier initialization
WARNING:tensorflow:From /usr/local/lib/python3.12/dist-packages/tensorflow/python/util/dispatch.py:1260: calling dropout (from tensorflow.pytho
Instructions for updating:
Please use `rate` instead of `keep_prob`. Rate should be set to `rate = 1 - keep_prob`.
I0000 00:00:1774983622.024903   33471 gpu_device.cc:2019] Created device /job:localhost/replica:0/task:0/device:GPU:0 with 13757 MB memory:  ->
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1774983622.293818   33471 mlir_graph_optimization_pass.cc:425] MLIR V1 optimization pass is not enabled
without pretraining.
/content/neural_graph_collaborative_filtering/NGCF/NGCF.py:484: DeprecationWarning: Conversion of an array with ndim > 0 to a scalar is depreca
  perf_str = 'Epoch %d [%.1fs]: train==[%.5f=%.5f + %.5f]' % (
Epoch 0 [231.6s]: train==[572.91626=572.61365 + 0.00000]
Epoch 1 [227.5s]: train==[354.94370=354.29645 + 0.00000]
Epoch 2 [227.8s]: train==[310.90353=310.00031 + 0.00000]
Epoch 3 [227.7s]: train==[293.04190=291.91757 + 0.00000]
Epoch 4 [227.2s]: train==[288.00735=286.68146 + 0.00000]
Epoch 5 [227.0s]: train==[282.16638=280.65143 + 0.00000]
Epoch 6 [227.5s]: train==[280.62946=278.93939 + 0.00000]
Epoch 7 [227.0s]: train==[281.33011=279.47345 + 0.00000]
Epoch 8 [228.0s]: train==[278.57181=276.55750 + 0.00000]
/content/neural_graph_collaborative_filtering/NGCF/NGCF.py:502: DeprecationWarning: Conversion of an array with ndim > 0 to a scalar is depreca
  perf_str = 'Epoch %d [%.1fs + %.1fs]: train==[%.5f=%.5f + %.5f + %.5f], recall=[%.5f, %.5f], ' \
Epoch 9 [227.2s + 1634.6s]: train==[283.72079=281.55414 + 2.16693 + 0.00000], recall=[0.01307, 0.04581], precision=[0.00606, 0.00441], hit=[0.1
save the weights in path:  /content/drive/MyDrive/NGCF_weights/weights/amazon-book/ngcf_norm_ngcf_l1/64/l0.01_r1e-05
Epoch 10 [227.3s]: train==[284.84628=282.53415 + 0.00000]
Epoch 11 [226.2s]: train==[282.80338=280.34924 + 0.00000]
Epoch 12 [226.2s]: train==[286.22672=283.63721 + 0.00000]
Epoch 13 [227.1s]: train==[285.57565=282.85355 + 0.00000]
```

```
Epoch 14 [226.0s]: train==[288.51904=285.66919 + 0.00000]
Epoch 15 [226.5s]: train==[289.38550=286.41345 + 0.00000]
Epoch 16 [227.1s]: train==[288.72025=285.63028 + 0.00000]
Epoch 17 [226.5s]: train==[287.27667=284.06973 + 0.00000]
Epoch 18 [226.6s]: train==[293.29868=289.97729 + 0.00000]
Epoch 19 [226.1s + 1653.4s]: train==[289.61600=286.18414 + 3.43161 + 0.00000], recall=[0.01419, 0.04842], precision=[0.00643, 0.00463], hit=[0.
WARNING:tensorflow:From /usr/local/lib/python3.12/dist-packages/tensorflow/python/training/saver.py:1068: remove_checkpoint (from tensorflow.py
Instructions for updating:
Use standard file APIs to delete files with this prefix.
save the weights in path:  /content/drive/MyDrive/NGCF_weights/weights/amazon-book/ngcf_norm_ngcf_l1/64/l0.01_r1e-05
Best Iter=[1]@[7838.4]  recall=[0.01419 0.02426 0.03324 0.04118 0.04842], precision=[0.00643    0.00565 0.00521 0.00489 0.00463], hit=[0.11096
```

Start coding or generate with AI.

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